Route 53 and DNS Basics

Explore the fundamentals of DNS and AWS Route 53 to understand domain resolution in cloud applications. Learn how to manage hosted zones and DNS records to route traffic effectively for public and private AWS resources, enabling you to build resilient and scalable applications.

Working with AWS, understanding DNS and how Route 53 handles domain resolution is foundational to building resilient, user-friendly applications. **DNS (domain name system)** translates human-readable domain names like `www.example.com` into IP addresses like `192.0.2.44` that machines use to communicate.

In AWS, **Route 53** is a scalable DNS web service that integrates tightly with the cloud infrastructure. As developers, we can register domains, manage DNS records, and route user requests with logic tailored to our applications. Let's understand what this means in detail.

Understanding DNS resolution

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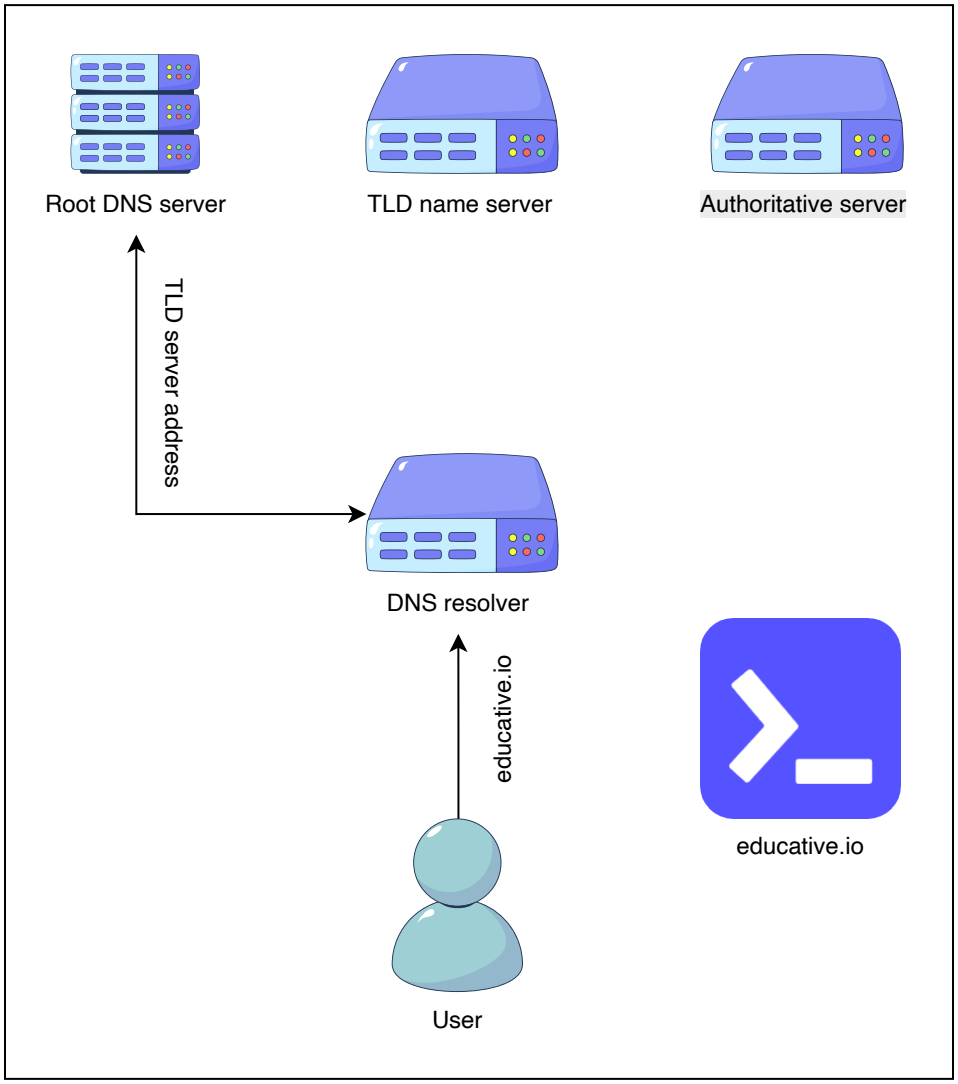
Before explaining how DNS resolution works, it's important to understand the role of a DNS server. A **DNS server** responds to queries about domain names and

their associated IP addresses. Different types of DNS servers are involved in the resolution process, including recursive resolvers, root name servers, top-level domain (TLD) servers, and authoritative name servers.



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Resolver and Root DNS

Route 53 acts as an authoritative DNS server, meaning it provides the final answer to DNS queries based on records configured within the hosted zones.

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DNS zones and records

A **DNS zone** is a defined section of the DNS namespace that a DNS server manages. You can think of it as the configuration space where all the records for a domain are stored, controlling how traffic is routed to resources like websites or services. In AWS, a DNS zone is created and managed using Route 53 as a hosted zone. It contains authoritative records that tell the DNS resolver how to handle requests for domain names. We can think of a hosted zone as the AWS-native implementation of a DNS zone, optimized for integration with other AWS services.

When we create a hosted zone in Route 53, we are defining a set of authoritative DNS records for a domain and its subdomains. Each DNS zone corresponds to a domain and can include multiple **resource record sets** that define how DNS queries should be answered.

Key DNS record types include:

DNS records

Record Type	Value
SOA records	Provides authoritative information about a DNS zone, includ name server, the zone administrator's email address, and ot parameters.
NS records	Specifies the authoritative name servers for a domain. Thes for providing DNS information about the domain.
A records	Maps a domain name to an IPv4 address. For example, we that maps "example.com" to "192.0.2.1".
AAAA records	Maps a domain name to an IPv6 address. For exa record that maps "example.com" to "2001:db8:33

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CNAME records	Maps one domain name to another. For instance, we might point “www.example.com” to “example.com”.
MX records	Specifies mail servers responsible for receiving email on bel records often point to mail servers like “mail.example.com”.
TXT records	Stores arbitrary text information associated with a domain r various purposes, such as verifying domain ownership or pr Policy Framework) records for email authentication.

Route 53 automatically creates a default NS and SOA (Start of Authority) record when a hosted zone is created. We typically don’t need to modify these unless working with complex delegation scenarios.

A **hosted zone** in Route 53 is a container for records that define how to route traffic for a domain. This zone is where we manage all the DNS records needed to direct traffic to your application or website.

We can create two types:

- **Public hosted zones** route traffic from the Internet to AWS resources.
- **Private hosted zones** are used within a VPC to resolve names for internal services.

How to configure Route 53

Before we can route traffic with Route 53, we first need a domain name. We can register a domain directly through the Route 53 console in a few simple steps:

1. Choose a unique domain name and confirm it’s available. If it's already taken, try different variations or a different top-level domain (TLD) like `.dev`, or `.cloud`.
2. Register the domain through Route 53 by providing the contact details.

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Behind the scenes, Route 53 sends your information to the registrar (Amazon Registrar or Gandi), who then submits it to the domain registry and WHOIS database.

Once registered, Route 53 automatically creates a *public hosted zone* with the same name as the domain and assigns four authoritative name servers. The service updates the domain's configuration to use these name servers so that Route 53 can respond to DNS queries.

Each **record** in the hosted zone tells Route 53 how to handle DNS queries:

- **Name:** This is the domain or subdomain (e.g., `www.educative.io`) you want to route traffic for. Route 53 automatically appends the hosted zone name.
- **Type:** Indicates the resource type, like `A` for IPv4 addresses or `MX` for mail servers.
- **Value:** Depends on the type. It could be an IP address for an `A` record or a domain name for a `CNAME`.

We can also create *alias records* to route traffic directly to AWS resources such as S3 buckets, CloudFront distributions, or load balancers.

After configuration, Route 53 follows the steps given below to fetch the resources:

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How Route53 works in DNS resolution

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